

Assignment No. 1

Subject: Database Engineering Lab (7CS271)

Class: SY CSE, AY 2025–26, Sem-II

Title: *Study of Commercial and Open-Source Database Management Systems*

CO: CO1

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1. Introduction

A Database Management System (DBMS) is software that enables users to efficiently store, retrieve, manage, and secure data. Modern DBMS solutions are broadly classified into **commercial (proprietary)** and **open-source** systems. This assignment studies the **architecture, features, use cases, and comparative aspects** of both categories.

2. Study of Commercial DBMS

2.1 Oracle Database

- **Architecture:** Multi-tier architecture with instance (SGA + background processes) and database (data files, redo logs).
- **Features:** High availability (RAC), advanced security, PL/SQL, data replication.
- **Use Cases:** Banking, telecom, large enterprise ERP systems.

2.2 Microsoft SQL Server

- **Architecture:** Client-server model with relational engine and storage engine.
- **Features:** T-SQL, integration with Microsoft ecosystem, strong BI tools.
- **Use Cases:** Enterprise applications, financial systems, .NET-based solutions.

2.3 IBM DB2

- **Architecture:** Distributed relational DBMS with shared-nothing architecture.
- **Features:** High performance analytics, strong transaction handling.
- **Use Cases:** Banking, insurance, data warehousing.

2.4 SAP HANA

- **Architecture:** In-memory, column-oriented database.
- **Features:** Real-time analytics, hybrid transactional/analytical processing (HTAP).
- **Use Cases:** ERP systems, real-time business intelligence.

3. Study of Open-Source DBMS

3.1 MySQL

- **Architecture:** Client-server architecture with pluggable storage engines.
- **Features:** Simplicity, high performance for read-heavy workloads.
- **Use Cases:** Web applications, CS platforms, startups.

3.2 PostgreSQL

- **Architecture:** Object-relational DBMS with extensible engine.
- **Features:** ACID compliance, complex queries, JSON support.
- **Use Cases:** Enterprise-grade open-source applications, analytics.

3.3 MongoDB

- **Architecture:** Document-oriented NoSQL database.
- **Features:** Schema-less design, horizontal scalability, JSON-like documents.
- **Use Cases:** Big data, IoT, real-time applications.

3.4 MariaDB

- **Architecture:** Fork of MySQL with enhanced performance.
- **Features:** Open governance, improved storage engines.
- **Use Cases:** Replacement for MySQL in enterprise systems.

4. Comparative Analysis:

Parameter	Commercial DBMS	Open-Source DBMS
Architecture	Complex, enterprise-grade, optimized for large workloads	Lightweight to extensible architectures
Scalability	Vertical + horizontal scaling	Mostly horizontal scaling
Security	Advanced built-in security features	Strong security but depends on configuration
Performance	Optimized for high-end enterprise use	Good performance, varies by DBMS
Cost	Expensive licenses	Free or low-cost
Customization	Limited (vendor-controlled)	High customization
Support	Vendor support	Community + optional paid support

5. Licensing Comparison:

DBMS Type	Licensing	DBMS Type	Data Model & Primary Use Case
Oracle	Proprietary, per-core licensing	Oracle	Relational / Large-scale Enterprise Apps, Mission-Critical
MS SQL Server	Proprietary, subscription-based	MS SQL Server	Relational / Windows Ecosystem Integration, Business Intelligence
IBM DB2	Proprietary	IBM DB2	Relational / Mainframe, Financial & Transaction Processing
SAP HANA	Proprietary	SAP HANA	In-Memory Relational / Real-time HTAP (Transactional & Analytical)
MySQL	GPL / Commercial dual license	MySQL	Relational / Web Applications, Content Management (LAMP stack)
PostgreSQL	PostgreSQL License (permissive)	PostgreSQL	Object-Relational / Complex Queries, GIS & Spatial Data
MongoDB	SSPL	MongoDB	Document (NoSQL) / Schema-less & Unstructured Data, Rapid Prototyping
MariaDB	GPL	MariaDB	Relational / Scalable Web Apps, Cluster-aware (MySQL fork)

6. References

Enterprise RDBMS Group:

- Oracle Database: <https://docs.oracle.com/en/database/>
- Microsoft SQL Server: <https://learn.microsoft.com/en-us/sql/sql-server/>
- IBM DB2: <https://www.ibm.com/docs/en/db2>
- SAP HANA: https://help.sap.com/docs/SAP_HANA_PLATFORM

Open-Source Databases Group:

- MySQL: <https://dev.mysql.com/doc/>
- PostgreSQL: <https://www.postgresql.org/docs/>
- MongoDB: <https://www.mongodb.com/docs/>
- MariaDB: <https://mariadb.com/kb/en/documentation/>

7. Conclusion

Commercial DBMS provide **high reliability, security, and enterprise-grade support**, making them suitable for mission-critical applications. Open-source DBMS offer **flexibility, cost-effectiveness, and strong community support**, making them ideal for startups and scalable applications. The choice of DBMS depends on **application size, budget, scalability requirements, and security needs**.